

MATH 345 Skeleton Notes

Unit 4: Orthogonality and least squares

Section 4.3: Least squares as projection

© 2024–2026 Jacob Denson, Kaitlyn Phillipson, Sebastien Roch. Except where otherwise noted, this work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

Least squares is projection when the subspace is $\text{col}(A)$.

Orthogonal projection can also be read as an optimization problem.

Given a subspace $U \subseteq \mathbb{R}^n$ and a vector $\mathbf{x}_0$, the projection $\text{proj}_U(\mathbf{x}_0)$ is the vector $\mathbf{x}^* \in U$ that minimizes $\|\mathbf{x} - \mathbf{x}_0\|$.

This is the first version of a pattern that will recur in Units 5 and 7: choose a space of allowable approximations, choose a way to measure error, and find the point in the allowable space with smallest error.

The equation $A\mathbf{x} = \mathbf{b}$ may have no solution because $\mathbf{b} \notin \text{col}(A)$. The least-squares problem asks for $\mathbf{x}$ so that $A\mathbf{x}$ is as close as possible to $\mathbf{b}$:

$$\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|.$$

Unit 2 asked whether a target vector was reachable. Least squares asks for the closest reachable output when the target is not reachable.

Note: Column spaces and null spaces Let A be an $m \times n$ matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Then

$$\text{col}(A)^\perp = \text{null}(A^T).$$

Indeed, a vector $\mathbf{r}$ is in $\text{col}(A)^\perp$ exactly when it is orthogonal to every column of A . That means

$$\mathbf{a}_1 \cdot \mathbf{r} = 0, \dots, \mathbf{a}_n \cdot \mathbf{r} = 0,$$

which is exactly the matrix equation $A^T \mathbf{r} = \mathbf{0}$.

Geometrically, $A\mathbf{x}$ is a point in $\text{col}(A)$. Thus the best possible $A\hat{\mathbf{x}}$ is the projection of $\mathbf{b}$ onto the column space:

$$A\hat{\mathbf{x}} = \text{proj}_{\text{col}(A)}(\mathbf{b}).$$

The residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to every column of A . Equivalently, $\mathbf{r} \in \text{col}(A)^\perp = \text{null}(A^T)$, so

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}, \quad A^T A\hat{\mathbf{x}} = A^T \mathbf{b}.$$

Theorem: Normal equations from residual orthogonality If $\hat{\mathbf{x}}$ minimizes $\|A\mathbf{x} - \mathbf{b}\|$, then the residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to $\text{col}(A)$. Equivalently, $\mathbf{r} \in \text{col}(A)^\perp = \text{null}(A^T)$, so

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0},$$

and every least-squares solution satisfies the normal equations

$$A^T A\hat{\mathbf{x}} = A^T \mathbf{b}.$$

The closest point $A\hat{\mathbf{x}}$ in $\text{col}(A)$ is unique. The coefficient vector $\hat{\mathbf{x}}$ is unique when the columns of A are linearly independent. If the columns are dependent, several different coefficient vectors may produce the same closest point.

Activity: Why $A^T A$ is invertible for independent columns

Let A be an $m \times n$ matrix. Prove that A has linearly independent columns if and only if $A^T A$ is invertible.

Activity: Unit 2 target revisited

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

1. Explain why $A\mathbf{x} = \mathbf{b}$ has no solution.
2. Form $A^T A$ and $A^T \mathbf{b}$.
3. Solve the normal equations.
4. Compute $A\hat{\mathbf{x}}$ and $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$.
5. Check $A^T \mathbf{r} = \mathbf{0}$.

Fit $y = c_0 + c_1 t$ to $(0, 1)$, $(1, 2)$, and $(2, 2)$. Then

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

The normal equations are $A^T A \mathbf{c} = A^T \mathbf{b}$. Since

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \quad A^T \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix},$$

we get $c_0 = 7/6$ and $c_1 = 1/2$.

The fitted line is $y = 7/6 + (1/2)t$. The fitted values are

$$A\mathbf{c} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix},$$

so the residual is

$$\mathbf{r} = \mathbf{b} - A\mathbf{c} = \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix}.$$

Check the orthogonality condition:

$$A^T \mathbf{r} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} -1/6 \\ 1/3 \\ -1/6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Code snapshot: fitting and checking the residual

```
xhat = np.linalg.lstsq(A, b, rcond=None)[0]
r = b - A @ xhat
A.T @ r
```

The vector $\mathbf{r}$ is the residual. The output of $A.T @ \mathbf{r}$ should be close to zero because the least-squares residual is orthogonal to the columns of A . QR will compute the same least-squares solution without directly forming $A^T A$.